CE 211: Thermal & Energy Sciences in Civil Engineering

Fall 2025, MWF: 14:30 – 15:20, WALC 1018

Instructor: Dr. Brandon E. Boor, Associate Professor of Civil & Construction Engineering
Office: HAMP G241
E-mail: bboor@purdue.edu
Office Hours: M – 15:30 to 17:00

Teaching Assistant: Chunxu Huang, Ph.D. Student
Office: HAMP G293A
E-mail: huan1445@purdue.edu
Office Hours: TBD

Teaching Assistant: Rebecca Liu, Ph.D. Student
Office: HAMP G293A
E-mail: liu3502@purdue.edu
Office Hours: TBD

Textbook (Required): Moran, M.J., Shapiro, H.N., Boettner, D.D., and Bailey, M.B. (2018). *Fundamentals of Engineering Thermodynamics, Enhanced eText, 9th Edition*. John Wiley & Sons, Inc, ISBN-13:978-1-119-39138-8. Available as eText via *VitalSource Bookshelf* app or through *WileyPLUS*. The hard copy of the 9th edition is also acceptable.

Course Materials: Purdue University Brightspace.

Course Objectives
The objectives of this course are: (1.) to introduce fundamental concepts of properties of materials, work, heat, internal energy, entropy, equilibrium, and relations derived from the first and second laws of thermodynamics and (2.) to learn how to apply these principles towards civil and construction engineering applications.

This course includes applications of the thermal sciences and energy fundamentals to civil and construction engineering topics. Emphasis is placed on fundamental concepts of properties of materials, work, heat, internal energy, entropy, equilibrium, and relations derived from the first and second laws of thermodynamics. Example applications include: renewable energy production; power plants; fluid flow in ducts/pipes; thermal properties of building/construction materials and processes; geothermal systems; heating, ventilation, and air conditioning (HVAC) processes in buildings; energy balances in buildings; refrigeration; contaminant transport in air, water, and soil; the urban heat island effect; and energy use in the transportation sector.

Learning Outcomes
Upon completion of this course, students will be able to:
1. Explain the principles of mass and energy conservation and their use in civil engineering applications.
2. Evaluate thermodynamic systems and several forms of work, energy, and heat transfer.
3. Apply conservation of mass and energy for closed systems and control volume processes.
4. Explain the second law of thermodynamics, entropy changes, and device efficiencies.

The learning process includes:
1. **Reading:** the textbook is a comprehensive source of information on fundamental thermal science topics.
2. **Lectures:** supplement fundamental concepts from the readings. Example problems will emphasize applications of thermal/energy science concepts in civil and construction engineering.
3. **Homework Assignments:** designed to reinforce the concepts presented during lectures. Students shall become familiar with computations for evaluating thermodynamic systems and retrieving thermodynamic property data.
4. **Quizzes and Exams:** intended to evaluate comprehension of topics covered in homework assignments.

Grading
The overall course grade will be weighted as follows:
Mid-Term Exam 1: 20% (October 02, 2025, 20:00 – 22:00, LILY 1105)
Mid-Term Exam 2: 20% (November 13, 2025, 20:00 – 22:00, WALC 1055)
Final Exam: 20% (Date/Time/Location TBD)
Quizzes: 25%
Homework Assignments: 10%
Attendance: 5%

The plus/minus grading system will be used (e.g. 96.7% and up = A+; 93.3% to 96.7% = A; 90.0% to 93.3% = A-; 86.7% to 90.0% = B+; 83.3% to 86.7% = B; 80.0% to 83.3% = B-). Modifications to this breakdown will be made at the instructor's discretion.

Mid-Term and Final Exams
There will be two in-person mid-term exams and an in-person final exam. No make-ups will be allowed unless arrangements have been made prior to the scheduled time. Make-up exams will be given only in the event of a verified emergency or doctor-verified sickness. Crib sheets will not be allowed on exams, but appropriate equation sheets will be supplied. This same policy will apply to the in-class quizzes. The final exam date, time, and location will be announced.

Quizzes
Quizzes will be given in class on a regular basis. The quizzes will closely mirror homework assignments and are intended to help students and the instructor identify topics that require further clarification. Quizzes will be announced at least one class period in advance.

Homework Assignments
Homework will be assigned approximately every week. Assignments should be done neatly on engineering quadrille paper and submitted to Brightspace as a PDF. All calculations must be shown to get full credit. Every number representing a physical quantity must have units with it. Final solutions should be clearly summarized and marked with a clear box around the answer. Assignments are to be turned in prior to the beginning of the lecture on the date due. Late assignments will not be accepted unless a specific arrangement has been made with the instructor at least a day prior to the due date. Put your name on everything you hand in.

Attendance
This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to the instructor is not possible, the student should contact the instructor as soon as possible by e-mail. When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond your control, and in cases of bereavement, quarantine, or isolation, the student or the student's representative should contact or go to the Office of the Dean of Students website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted for cases of grief/bereavement, military service, jury duty, and parenting leave. For details, see the Academic Regulations & Student Conduct section of the University Catalog website. The use of cell phones during lecture is prohibited. Tablets or laptops with a stylus (e.g. iPad Pro or Microsoft Surface) are permitted for notetaking. Fall 2025 semester holidays (no lecture): Labor Day (September 01, 2025), Fall Break (October 13-14, 2025), and Thanksgiving Break (November 24-28, 2025).

Lecture Notes
Basic lecture PDFs will be prepared and uploaded to Brightspace prior to each class together with blank space for hand-written notes given in real-time by the instructor using an iPad Pro. You are responsible for printing

and bringing the lecture PDFs to class and documenting all hand-written notes by the instructor during lecture. It is recommended that you print the notes (1 slide per page) ahead of class or use a tablet with a stylus for note-taking (PDF Expert or similar is recommended).

E-mail Communication
All e-mails directed to the instructor and teaching assistants must be written in a professional manner.

Prerequisites
CHM 11500 with a minimum grade of C- *OR*
CHM 12300 with a minimum grade of C- *OR*
CHM 11100 with a minimum grade of C- and CHM 11200 with a minimum grade of C- **AND**
MA 26100 with a minimum grade of C- (may be taken concurrently) *OR*
MA 26300 with a minimum grade of C- (may be taken concurrently) *OR*
MA 18200 with a minimum grade of C- (may be taken concurrently) *OR*
MA 27100 with a minimum grade of C- (may be taken concurrently)

Course Evaluation
During the last two weeks of the semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system. You will receive an official e-mail from evaluation administrators with a link to the online evaluation site. You will have up to two weeks to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Academic Integrity
Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either e-mailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty" [Part 5, Section III-B-2-a, University Regulations]. Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest" [University Senate Document 72-18, December 15, 1972].

Purdue Honors Pledge
As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do.
Accountable together - we are Purdue.

Grief Absence Policy for Students
Purdue University recognizes that a time of bereavement is very difficult for a student. The University therefore provides the following rights to students facing the loss of a family member through the Grief Absence Policy for Students (GAPS). GAPS Policy: students will be excused for funeral leave and given the opportunity to earn equivalent credit and to demonstrate evidence of meeting the learning outcomes for missing assignments or assessments in the event of the death of a member of the student's family.

Counseling and Psychological Services (CAPS)
Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 and

http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Students with Disabilities
Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let the instructor know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Emergencies
In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructor via e-mail. You are expected to read your @purdue.edu e-mail on a frequent basis.

Diversity and Inclusion
Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services, and activities consistent with applicable federal, state, and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Executive Memorandum No. D-1, which provides specific contractual rights and remedies. Any student who believes they have been discriminated against may visit http://www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

CE 211 Lecture Schedule – Fall 2025 (Revision: August 21, 2025)

Lecture	Date	Topics Covered	Reading
1	Aug. 25	Course introduction.	1.1-1.2
2	Aug. 27	Introduction to thermodynamics terminology.	1.3-1.5
3	Aug. 29	Processes, properties, and units.	1.6-1.9, 2.1
-	Sept. 01	**No lecture:** Labor Day.	-
4	Sept. 03 & Sept. 05	Mechanical concepts of energy.	2.2-2.4
5	Sept. 08	Additional forms of work and energy.	2.5-2.7
6	Sept. 10 & Sept. 12	Heat transfer and energy balances.	-
7	Sept. 15	Work, heat transfer, and energy balance example problems.	-
8	Sept. 17	Thermodynamic cycles.	3.1-3.3
9	Sept. 19	Evaluating thermodynamic properties I.	3.3-3.5
10	Sept. 22	Evaluating thermodynamic properties II.	-
11	Sept. 24	Evaluating thermodynamic properties: example problems.	3.6-3.11
12	Sept. 26 & Sept. 29	Solving closed system energy balances.	-
13	Oct. 01	Practice problems for Mid-Term Exam 1.	-
-	Oct. 02	**Mid-Term Exam 1: 20:00 – 22:00, LILY 1105**	
14	Oct. 03 & Oct. 06	Specific heats.	3.12-3.15
15	Oct. 08	Ideal gases.	-
16	Oct. 10	Conservation of mass for control volumes I.	4.1-4.3
-	Oct. 13	**No lecture:** Fall Break	-
17	Oct. 15	Conservation of mass for control volumes II.	-
18	Oct. 17 & Oct. 20	Conservation of energy for control volumes.	4.4-4.7
19	Oct. 22	Analyzing control volumes at steady state I.	4.8-4.10
20	Oct. 24	Analyzing control volumes at steady state II.	4.11-4.12
21	Oct. 27	Transient control volumes and system integration.	-
22	Oct. 29	Control volume example problems.	-
23	Oct. 31 & Nov. 03	Introduction to ideal gas mixtures and psychrometrics.	12.1-12.3
24	Nov. 05 & Nov. 07	Psychrometric parameters and psychrometric chart.	12.4-12.7
25	Nov. 10 & Nov. 12	Practice problems for Mid-Term Exam 2.	-
-	Nov. 13	**Mid-Term Exam 2: 20:00 – 22:00, WALC 1055**	
26	Nov. 14 & Nov. 17	Psychrometric processes and dehumidification in HVAC systems.	12.8
27	Nov. 19	Humidification in HVAC systems.	12.9
28	Nov. 21	The 2nd law of thermodynamics I.	5.1-5.10
-	Nov. 24-28	**No lectures:** Thanksgiving Break	-
29	Dec. 01	The 2nd law of thermodynamics II.	5.10-5.11
30	Dec. 03	Entropy.	6.1-6.5
31	Dec. 05	The 2nd law of thermodynamics for closed systems and control volumes.	6.7-6.10
32	Dec. 08 & Dec. 10	Steady-state entropy balance, isentropic processes, and isentropic efficiencies.	6.11-6.12
33	Dec. 12	Practice problems for Final Exam.	-